

STUDENT ACADEMIC PERFORMANCE REPORT AND COURSE SYLLABUS

Student Information

Particulars	Details
Student Name	Gautham Manoj
Roll Number	AM.ID.R4IDS23007-FT
Academic Program & Branch	Ph.D. 2023
Current CGPA	8.76

Semester Performance Details

Semester 4

Semester	Course Code	Course Name	Academic Term Period	Course Type	Grade Obtained
4	22BIO811	Integrated Systems Genomics	2024-2025 Even Semester	Regular	A

Performance Summary

Metric	Score
SGPA	9.0
CGPA	8.76

Course Details

Course Code: 22BIO811

Course Title: Integrated Systems Genomics

Credits: 4

Instructor

Dr. Prashanth N. Suravajhala

Grade Obtained: A

COURSE OVERVIEW

Integrated Systems Genomics is an advanced interdisciplinary course that integrates Systems Biology, Functional Genomics, Bioinformatics, Next Generation Sequencing (NGS), and Non-Coding RNA Biology. The course introduces students to biological interaction networks, protein-protein interactions (PPIs), systems genomics approaches, genome-scale sequencing technologies, transcriptomics, metagenomics, and regulatory RNA molecules. The course combines theoretical concepts with practical hands-on exercises, visualizers, computational tools, and case studies to provide a comprehensive understanding of modern genomics research.

LEARNING OBJECTIVES

The course enables students to understand the interactions within biological networks and explore fundamental aspects of protein-protein interactions, their biological significance, and applications.

Students will gain knowledge of Systems Biology concepts and their transition into modern Next Generation Sequencing technologies, with emphasis on genome-wide analyses and characterization of non-coding RNAs.

The course also provides insights into the biology of regulatory RNA molecules, their transcription, gene regulation mechanisms, and their influence on cellular processes and gene expression.

COURSE SYLLABUS

UNIT I – SYSTEMS BIOLOGY AND PROTEIN-PROTEIN INTERACTIONS

Introduction to top-down and bottom-up Systems Biology approaches.

Analytical methods for understanding Protein-Protein Interactions (PPIs).

Tools and visualizers used for studying and analyzing Protein-Protein Interaction networks.

Regulatory Genomics and Systems Genomics.

Hands-on exercises and practical applications.

UNIT II – NEXT GENERATION SEQUENCING TECHNOLOGIES

Introduction to Next Generation Sequencing (NGS), including its significance and applications.

Concepts of Whole Exome Sequencing (WES) and Whole Genome Sequencing (WGS).

Concepts of Whole Transcriptome Shotgun Sequencing (WTSS).

RNA-Seq technology and transcriptome analysis.

UNIT III – MICROBIAL GENOMICS AND ADVANCED SEQUENCING

Introduction to Bacterial Genome Sequencing.

Fundamentals of Metagenomics and microbial community analysis.

Introduction to Oxford Nanopore Sequencing technology.

Introduction to Galaxy bioinformatics platform.

Case studies and practical applications.

UNIT IV – NON-CODING RNA BIOLOGY

Introduction to Non-Coding RNAs (ncRNAs), their significance and biological roles.

Small Non-Coding RNAs and Long Non-Coding RNAs.

Non-Coding RNAs as regulatory molecules.

miRNA-mRNA interactions.

lncRNA-protein interactions.

Tools and computational approaches for identifying ncRNAs.

Practical exercises and case studies.

REFERENCE MATERIALS

Reference 1

Lu, H., Zhou, Q., He, J. et al.

Recent Advances in the Development of Protein-Protein Interaction Modulators: Mechanisms and Clinical Trials.

Signal Transduction and Targeted Therapy, 5, 213 (2020).

Reference 2

Chuang HY, Hofree M, Ideker T.

A Decade of Systems Biology.

Annual Review of Cell and Developmental Biology, 2010;26:721–744.

Reference 3

Goodwin, S., McPherson, J., McCombie, W.

Coming of Age: Ten Years of Next-Generation Sequencing Technologies.

Nature Reviews Genetics, 17, 333–351 (2016).

Reference 4

Non-Coding RNA Collection

Cell Press Resource Collection on Non-Coding RNA.

COURSE OUTCOMES (COs)

Upon successful completion of the course, students will be able to:

CO1

Understand and identify Protein-Protein Interactions (PPIs) and determine biologically significant interacting partners for target proteins.

CO2

Compare computational (in silico) approaches with experimentally validated biological methods and understand best practices in Next Generation Sequencing analysis.

CO3

Perform validation strategies and identify candidate mutations and Differentially Expressed Genes (DEGs) from genomic and transcriptomic datasets.

CO4

Understand the biological significance of Non-Coding RNAs (ncRNAs), including long non-coding RNAs (lncRNAs), microRNAs (miRNAs), Piwi-interacting RNAs (piRNAs), and other regulatory RNA molecules.

CO5

Apply bioinformatics tools, visualizers, computational workflows, and analytical methods to solve real-world genomics and systems biology problems.

TEACHING AND LEARNING APPROACH

The course is designed to be highly interactive and self-contained with extensive brainstorming sessions.

Students will participate in:

- Interactive discussions
- Hands-on computational exercises

- Bioinformatics tool demonstrations
- Network visualization exercises
- Command-line based genomic analysis
- Case study evaluations
- Research-oriented problem-solving activities

COURSE RESULT SUMMARY

Course Code	Course Title	Semester	Credits	Grade	SGPA
22BIO811	Integrated Systems Genomics	4	4	A	9.0

ACADEMIC RECORD

Student Name: Gautham Manoj

Roll Number: AM.ID.R4IDS23007-FT

Programme: Ph.D. 2023

Course Completed: Integrated Systems Genomics (22BIO811)

Semester: 4

Grade Awarded: A

Semester GPA (SGPA): 9.0

Cumulative GPA (CGPA): 8.76

Academic Term: 2024–2025 Even Semester

Prepared as a consolidated academic record containing the student's performance details and complete syllabus for **22BIO811 – Integrated Systems Genomics**.